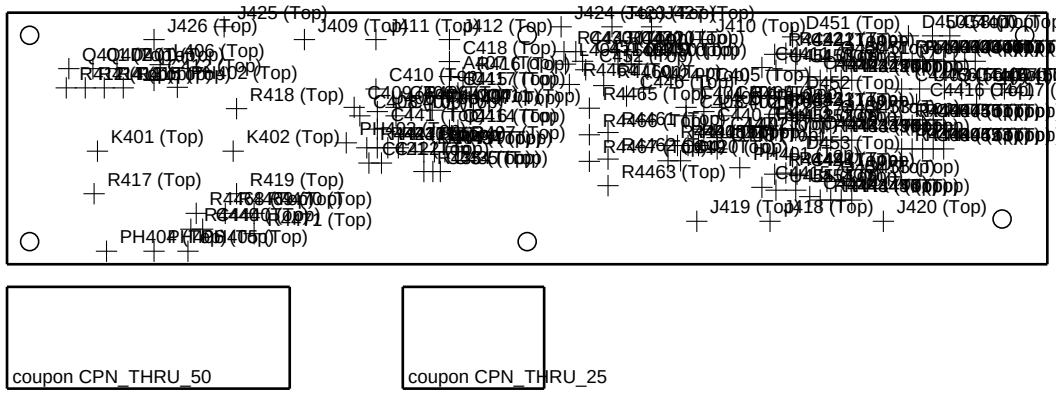

Board extent 184.000 mm (X) x 44.500 mm (Y) (without the 2 coupons)

Top view, Edge.Cuts and holes of RF.kicad_pcb; crosses = CPL positions (KiCad footprint origins). Not to a fixed scale.
NOT RELEASED: model/CI only; bench gates OPEN

RF fab notes

NOT RELEASED: model/CI only; bench gates OPEN.
Board status: RF2_ROUTED_NOT_FOR_FABRICATION. Every value below is a CI/model output of pcb/fab_export.py and pcb/board_gen.py; no hardware or bench gate is closed by it.

Board

Outline (Edge.Cuts of RF.kicad_pcb): 184.000 x 44.500 mm, area 8188.0 mm2; lower-left corner at X 100.000, Y -144.500 mm in the Gerber/CPL frame (KiCad absolute origin, Y up).
Holes: NPTH 3.20 mm, PTH 3.20 mm (6 mounting holes; NPTH unplated and without net).

Test coupons (separate pieces)

Edge.Cuts holds 3 separate closed outlines: the board and 2 impedance test coupons. The coupons are not joined to the board (no tab or mouse-bite), are not boards and get no assembly; make them with the board, same stack-up, and tell the fab at order time that the Gerber holds 3 pieces (the panelisation is an order option, OPEN).
Coupon CPN_THRU_50: X 100.000 to 150.000, Y -166.500 to -148.500 mm in the Gerber/CPL frame (board frame x 0 to 50, y -22 to -4 mm); an RF50 CPWG thru line like the board lines, own ground net CPN_GND_50; edge SMAs CPN1, CPN2 are not in the CPL or BOM (hand-fitted for the bench).
Coupon CPN_THRU_25: X 170.000 to 195.000, Y -166.500 to -148.500 mm in the Gerber/CPL frame (board frame x 70 to 95, y -22 to -4 mm); an RF50 CPWG thru line like the board lines, own ground net CPN_GND_25; edge SMAs CPN3, CPN4 are not in the CPL or BOM (hand-fitted for the bench).

Fabrication

Copper layers: 4; finished thickness 1.6 mm (JLC tolerance +/-10 % for boards >= 1.0 mm).
Stack-up JLC04161H-7628 (https://jlcpcb.com/impedance, read 2026-11-07; pcb/generated/netclasses.json stackup (pcb/stackup.py)):

- L1 F.Cu: copper 0.035 mm
- prepreg 7628x1: 0.2104 mm, Er 4.4
- L2 In1.Cu: copper 0.0152 mm
- core: 1.065 mm, Er 4.6
- L3 In2.Cu: copper 0.0152 mm
- prepreg 7628x1: 0.2104 mm, Er 4.4
- L4 B.Cu: copper 0.035 mm

Layer thicknesses sum to 1.5862 mm (computed). Copper weights: outer 0.035 mm = 1 oz, inner 0.0152 mm = 0.5 oz finished (JLC "H/HOZ").
Impedance: 50 ohm CPWG on L1 over solid L2, w 0.36 mm, gap 0.2 mm, +/-10 % (JLC standard): order WITH impedance control (44 RF50 nets).
The finished impedance is a coupon measurement (bench gate, OPEN); the design value is a MODEL.
Surface finish, solder mask and legend colours: order options, not fixed by

this fab set.
KiCad 10.0.7 DRC on the committed board: 0 errors, 0 unconnected, 398 reviewed warnings, ignored checks [] (report.json; CI/model result).

Files (fab/)

gerber/: Gerber X2 with netlist attributes, Protel extensions, one file per layer, Gerber job file (.gbrjob). drill/: Excellon PTH and NPTH files (mm, decimal, absolute origin, alternate oval mode) and Gerber X2 drill maps. RF.ipc356: IPC-D-356 netlist of every net with a pad. assembly/: JLC BOM and CPL, the raw KiCad position file, the rotation offset table and the instance map.

Assembly

Boards to build: 1 = instances (RF 1) + spares 0; every board is the same design and is assembled the same.
Sides with parts: Top (SMD 174, THT 3).
CPL: Mid X / Mid Y are the KiCad footprint origins (JLC defines them as the component centroid) and Rotation the raw KiCad rotation (degrees, counter-clockwise as the JLC CPL column). JLC rotation and centroid offsets per footprint are TBD (assembly/RF-cpl-rotation-offsets.csv): check every part in the JLC viewer at order time; nothing here verifies them.
BOM TBD: L402 (0805LS-471XGRC): TBD: build/10-jlc-bom.csv row has no LCSC code
BOM TBD: PH401 (A402 SAWbird+ NOAA OUT (E4_RF_CH1) board end, placeholder): TBD: pilot_placeholder, no graph ref and no MPN yet; replaced by GR-4 graph ref for this board end (cable assembly and length: SRC-2)
BOM TBD: PH402 (A405 VAT attenuator IN (E4_RF_CH5_DC) board end, placeholder): TBD: pilot_placeholder, no graph ref and no MPN yet; replaced by GR-4 graph ref for this board end (cable assembly and length: SRC-2)
BOM TBD: PH403 (A405 VAT attenuator OUT (E4_RF_CH5_ATT) board end, placeholder): TBD: pilot_placeholder, no graph ref and no MPN yet; replaced by GR-4 graph ref for this board end (cable assembly and length: SRC-2)
BOM TBD: PH404 (A406 Boni bias-tee RF port (E4_HF_BT) board end, placeholder): TBD: pilot_placeholder, no graph ref and no MPN yet; replaced by GR-4 graph ref for this board end (cable assembly and length: SRC-2)
BOM TBD: PH405 (W408 OUT / A406 ANT tap point (E4_HF_CM) board end, placeholder): TBD: pilot_placeholder, no graph ref and no MPN yet; replaced by GR-4 graph ref for this board end (cable assembly and length: SRC-2)
BOM TBD: PH406 (A414 Airspy HF+ ANT (E4_HFP_ANT) board end, placeholder): TBD: pilot_placeholder, no graph ref and no MPN yet; replaced by GR-4 graph ref for this board end (cable assembly and length: SRC-2)

Not proved here (OPEN)

- that the schematic is right
- current capacity or temperature rise beyond the chosen class widths
- impedance, RF loss or isolation, EMC or signal integrity
- creepage beyond the encoded clearances
- that the enclosure fits (only outline and hole checks; heights are not checked here)
- that the fab's DFM accepts the board, or that CPL rotations are right
- that the land patterns match the purchased MPNs
- any function: hardware and bench gates stay OPEN

Sources (read 2026-11-07): https://jlcpcb.com/help/article/bill-of-materials-for-pcb-assembly; https://jlcpcb.com/help/article/pick-place-file-for-pcb-assembly; https://jlcpcb.com/help/article/how-to-generate-the-bom-and-centroid-file-from-kicad; https://jlcpcb.com/help/article/how-to-generate-gerber-and-drill-files-in-kicad-8; https://jlcpcb.com/capabilities/pcb-capabilities